

EXPERIMENT
0 assumptions changed · Experiment equals Base

CHANGE
Controls are grouped by **what kind of thing** they are. They are not equivalent.

OPENING STATE
The engine already accepted this. Default is one age-0 €20m cohort and €10m cash — the v0.4 company. A vintage mix writes {arr, age} cohorts. Not an adapter; no file, no company name.

Opening ARR €20.00m
installed-base ARR at t=0 · base €20.00m

Opening cash €10.00m
cash balance at t=0 · base €10.00m

Early vintage (age 0) 100%
share of opening ARR in a brand-new cohort · base 100%

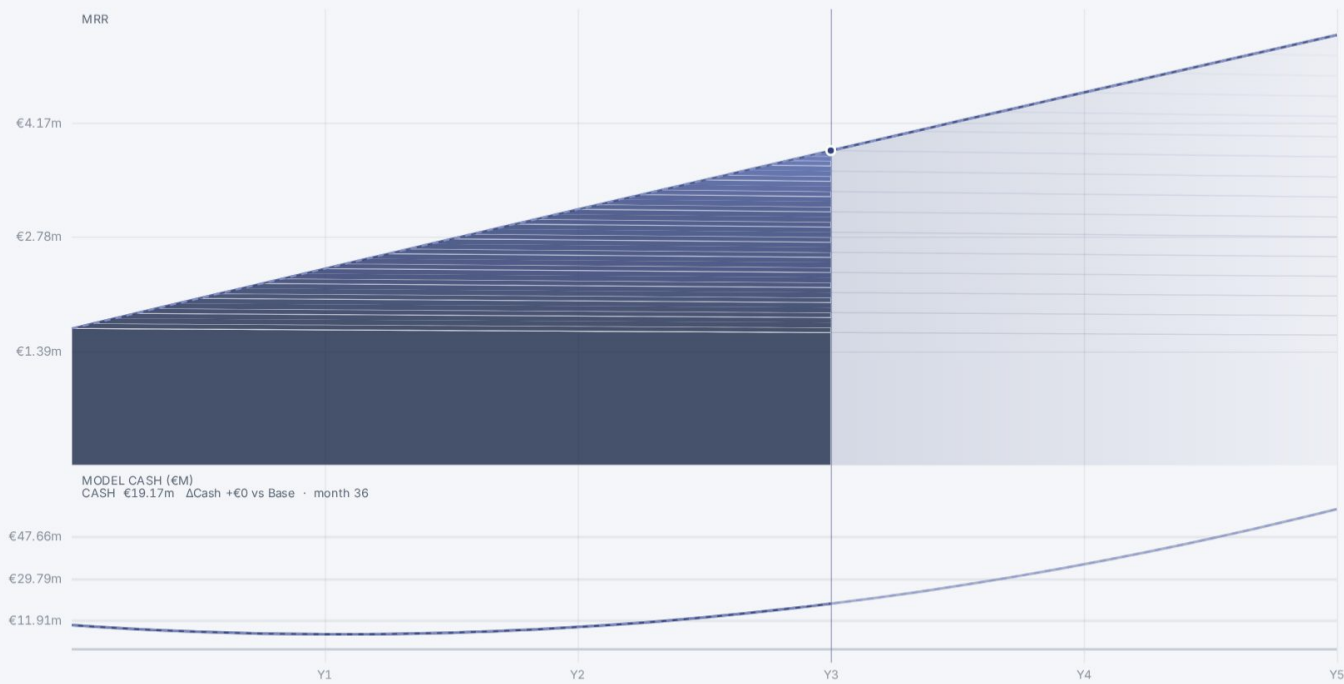
Developing vintage (age 12) 0%
share at the first band edge · base 0%

Mature vintage (age 24) 0%
share in the terminal band · base 0%

MANAGEMENT INPUT
S&M is a stated intention. Off by default the engine spends it in full even if cash is already negative. A cash reserve is the bound that stops that.

S&M investment €900k
Intended commercial investment per month · base €900k

S&M cash reserve off · unconstrained
S&M cannot spend through this cash floor. Off = unconstrained (prior). · base off · unconstrained



WHERE THE COMPANY IS · MRR
€3.83m on Base
Revenue run rate €45.66m
Gross margin 80.0%
Gross profit · this month €3.04m
EBITA / FCF · this month €1.09m
Cash €19.17m

HOW IT IS MOVING · MRR
Opening €3.78m
+ New €63k
+ Expansion €30k
- Leakage €33k
Closing €3.83m

INSTALLED-BASE NET · MRR
Expansion - Leakage -€3k
Net installed-base movement of -€3k against opening installed-base MRR of €3.78m = **0.08%** this month. The installed base is **net-shrinking** (tolerance ±0.05% of opening).

WHAT IT IS MADE OF · MRR
Opening base still here €1.62m · 42.2%
From cohorts acquired since €2.22m · 57.8%
Cohorts alive 37

EMERGENT KPIS
ARR growth, year on year 22.9%
R12M GRR 89.6%
R12M NRR 99.0%
CAC payback 18.0 months · Base 18.0
Measurements of the system, not inputs to it.

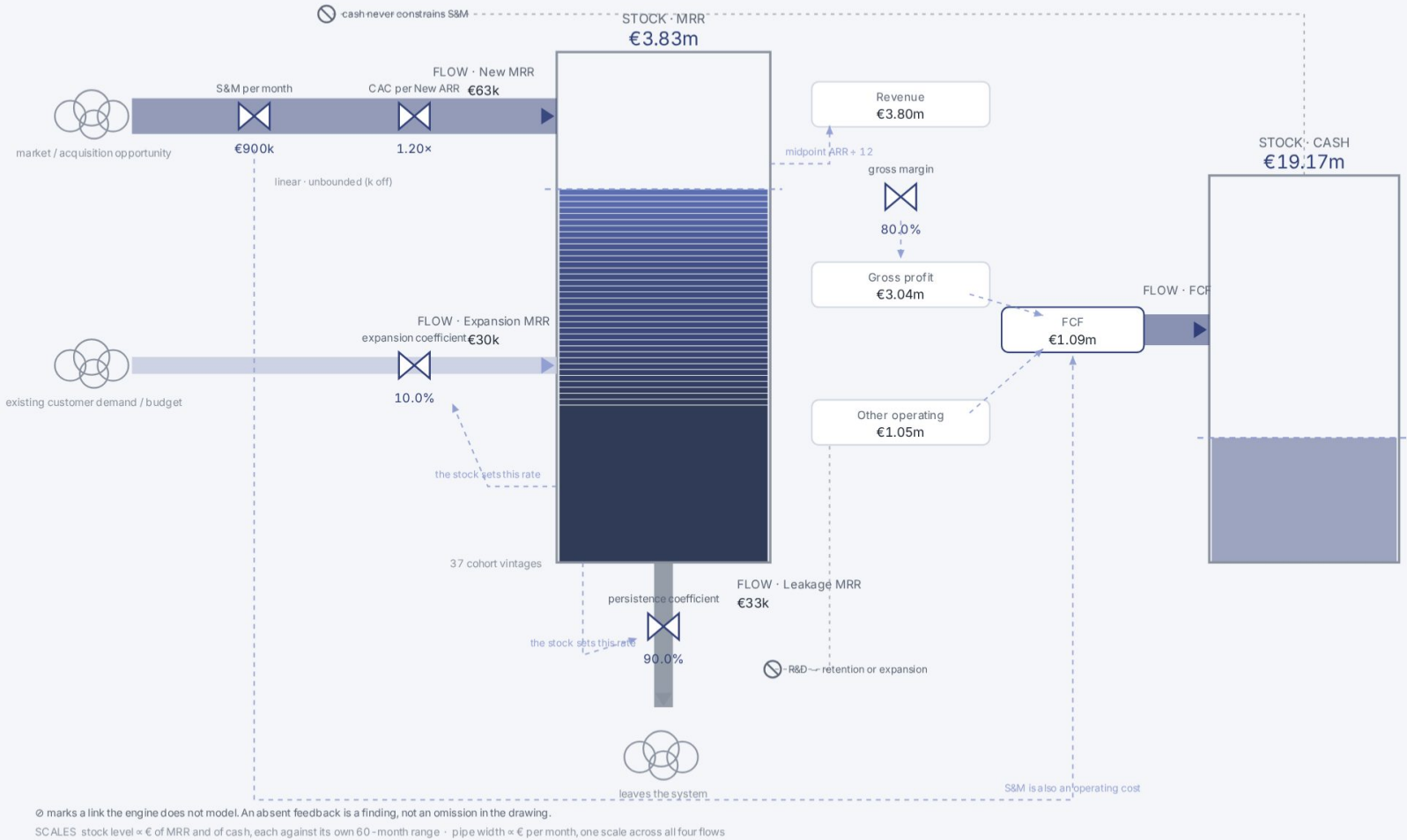
INSTALLED-BASE EVOLUTION · R12M
Opening cohort 100.0%
- Gross leakage impact -10.4%
+ Expansion impact +9.4%
= NRR 99.0%

Current v1 engine combines churn and contraction in Gross Leakage — it cannot yet say how much of the loss was lost logos versus shrinking accounts.

Experiment matches Base

SYSTEM · stock and flow · month 36

Rectangles accumulate. Pipes fill and drain them. Bowties are the valves you set. Clouds are the model boundary. Dashed indigo arrows are information: the stock setting the rate of its own future flows.



STOCKS · MONTH 36 · MRR

MRR	€3.83m
Cash	€19.17m

Two things accumulate in this business. Everything else is a rate.

FLOWS INTO AND OUT OF MRR

+ New MRR	€63k
+ Expansion MRR	€30k
- Leakage MRR	€33k
= net change in the stock	€59k

Net €59k this month against a stock of €3.83m — 1.5%. That ratio is why a stock this large is slow to turn, and why one month of flow could never be drawn at the stock's own scale.

THE STOCK SETS ITS OWN RATES

Leakage is $€3.78m \times 0.874\%$. Expansion is the retained balance $\times 0.797\%$. Revenue is the midpoint of the stock $\times 12$. Three of the four flows are functions of the stock they act on — the blue dashed arrows. Only New ARR is set from outside, by S&M — linearly at the default (saturation spend off).

TWO LINKS THE ENGINE DOES NOT HAVE

⊗ **Cash** → **S&M**. S&M is exogenous. Cash can fall to €6.10m and nothing throttles spend. The capital loop is drawn open because it is open. Set a cash reserve on the Forces rail to close it.

⊗ **R&D** → **retention or expansion**. R&D is a pure cost with no modelled benefit, so it reaches no valve. The absence is the honest drawing of the model boundary.

WHY THIS SURFACE EXISTS

This surface shows something a KPI table cannot: **which rates are functions of which stocks**, and which links do not exist. A fourth candidate — an intra-month bridge animation — established no unique percept and was rejected. It is preserved in the research archive, not shipped here.

Acquisition adds recurring state. The installed base expands or leaks. Finance records the consequence. A model audit, not a dashboard: the causal topology the engine actually contains, with ∅ marking the links it does not. Click any bowtie valve to set the assumption where it acts.

EXPERIMENT

Efficiency vs Spend pair — Base buys New ARR with acquisition efficiency, Experiment buys the same New ARR with S&M. See the Scenarios panel for the calibrated values.

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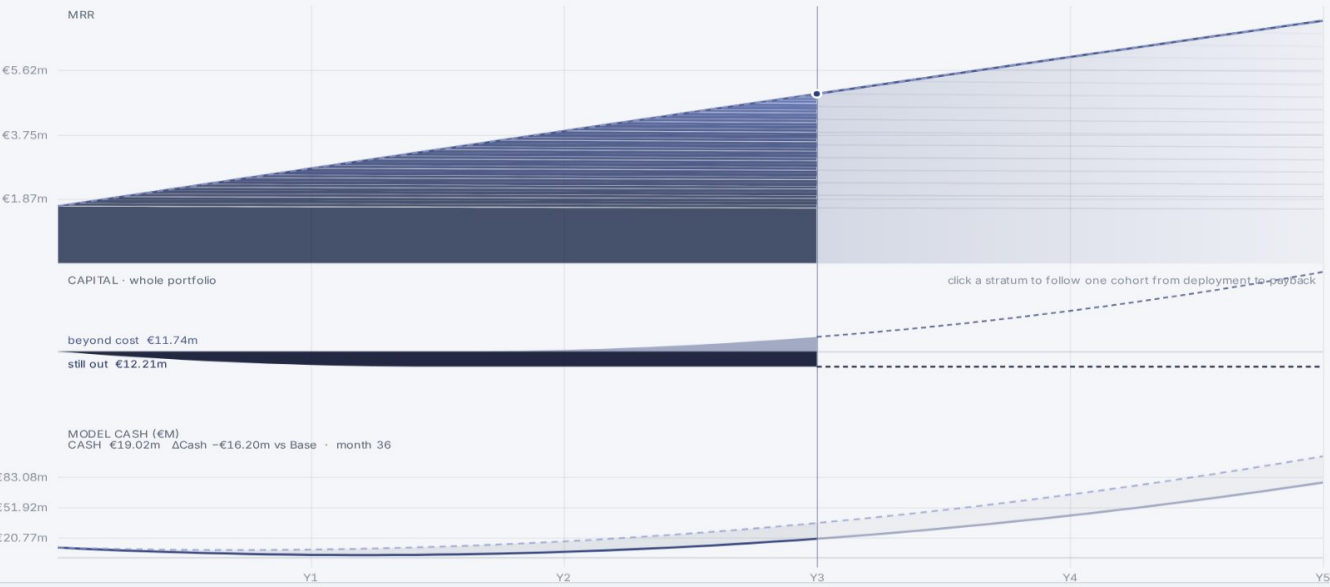
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share in the terminal band · base 0%

MANAGEMENT INPUT

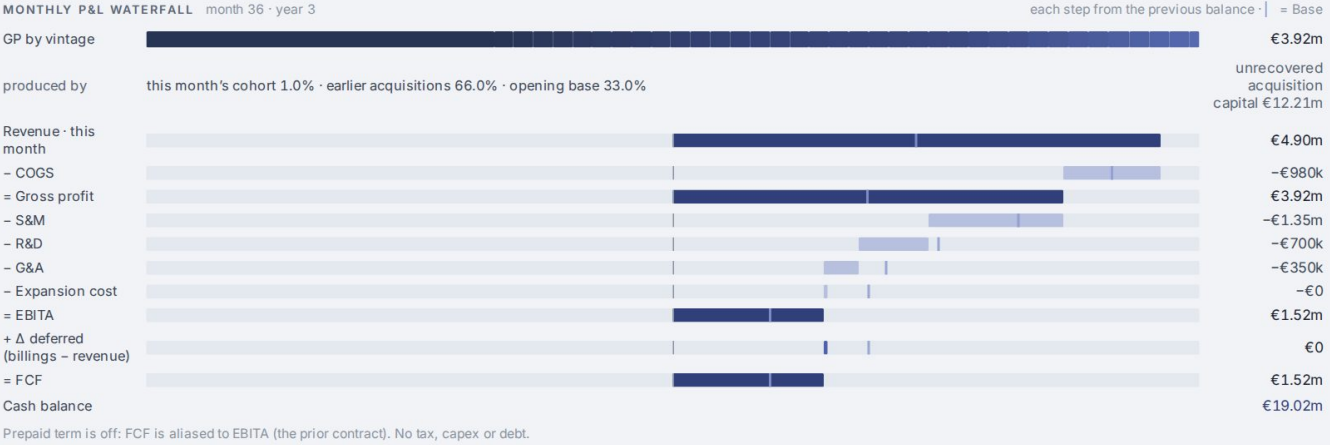
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S&M investment €1.35m
intended commercial investment per month · base €900k

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S&M cannot spend through this cash floor. Off = unconstrained (prior). · base off · unconstrained.



CANONICAL SCENARIOS 1 · Retention 2 · Expansion 3 · Acquisition efficiency 4 · Margin 5 · Efficiency vs Spend 6 · Same KPIs, different history Clear



SCENARIO 5 · EFFICIENCY VS SPEND

Same ARR trajectory, different capital journey

ARR alone does not reveal the amount of capital required to produce it.

CONSEQUENCE · EXPERIMENT - BASE

M60 MRR	+€0
New MRR per month	+€0
Acquisition capital deployed	+€27.00m
Cumulative FCF	-€27.00m
Ending cash	-€27.00m
CAC payback	12.0 → 18.0 months

The installed base stays net-shrinking under the model's transition laws ($g = 0.999163$ per month).

$g = (1-\ell)(1+e)$, the monthly multiplier applied to existing ARR before new business.

MODEL BOUNDARIES

EXPERIMENT
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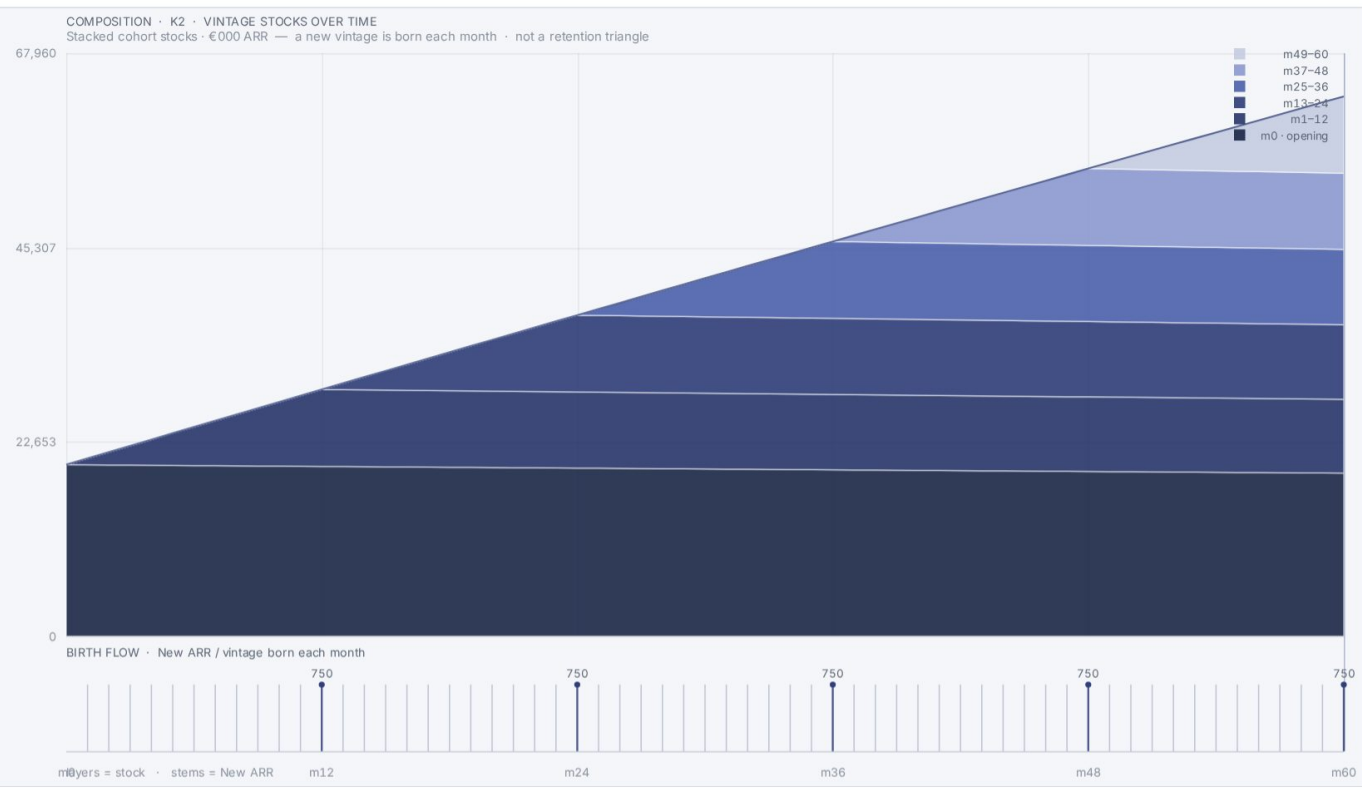
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COHORTS · VINTAGES / STOCKS / RATES · VS COMPANY (WHOLE-FIRM) · VS APPENDIX (MONTH GRID)
Which vintage is carrying/leaking the book — and is blended NRR lying?



VINTAGE READING · WHICH VINTAGE CARRIES / LEAKS				
△ blended NRR Blended NRR 99.0% is the vintage NRR. Tenure is flat — mix is a no-op, not a cohort story.				
m0 GRR 89.6% NRR 99.0% leaks the book expansion 9.4% · contraction —	m12 GRR 89.6% NRR 99.0% leaks the book expansion 9.4% · contraction —	m24 GRR 89.6% NRR 99.0% leaks the book expansion 9.4% · contraction —	m36 GRR 89.6% NRR 99.0% leaks the book expansion 9.4% · contraction —	m48 GRR 89.6% NRR 99.0% leaks the book expansion 9.4% · contraction —
Logos ≠ ARR. Logo count is a customer stock; ARR is the recurring-state stock. · NRR – GRR = expansion (measured). Contraction is — while the logo bound is off — leakage is one number.				
Flat-law vintage mix is a no-op — do not claim the stack. Tenure / Scenario 6 give age a rate. · <u>Tenure is flat — open Scenario 6 / Tenure</u>				

COHORTS
Which vintage is carrying/leaking the book — and is blended NRR lying? Composition is yearly vintage stock. Age is a cross-section. Blended NRR is a warning, not a tile.

THIS WORLD	
Reading	Base
Mode	Composition
Month 60 ARR · €000	62,926
R12M GRR	89.6%
R12M NRR	99.0%
Blended NRR 99.0% is the vintage NRR. Tenure is flat — mix is a no-op, not a cohort story.	

VINTAGE SHARE OF THE BOOK	
m0 share of book	30.2%
m12 share of book	1.1%
m24 share of book	1.2%
m36 share of book	1.2%
m48 share of book	1.2%
Share of closing ARR at the selected month. Carry / leak is retained vs that vintage's own initial ARR.	

ENGINE AGE BANDS	
Early · engine band	8,959 · 14.2%
Developing · engine band	8,869 · 14.1%
Mature · engine band	45,098 · 71.7%
Bars are ARR. Logo counts are — while the logo bound is off. Display bars on the stage are 12-month buckets of the same ages; the engine still has three laws.	

Scenario 6 · same KPIs, different history
Tenure laws on the rail write the same three-band array. No new coefficient.

SaaS Physics

an economic system, observed

base vs experiment

Company

System

Scenarios

Cohorts

Appendix

BASIS

MRR

ARR

Method

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Opening ARR

€20.00m

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Opening cash

€10.00m

cash balance at t=0 · base €10.00m

Early vintage (age 0)

100%

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Developing vintage (age 12)

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share at the first band edge · base 0%

Mature vintage (age 24)

0%

share in the terminal band · base 0%

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S&M investment

€900k

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S&M cash reserve

off · unconstrained

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COHORTS · VINTAGES / STOCKS / RATES · VS COMPANY (WHOLE-FIRM) · VS APPENDIX (MONTH GRID)

Vintage stock layers

Birth flow · new vintage each month

Vintage reading always on

€000

Base

Experiment

Composition

Age profile

Which vintage is carrying/leaking the book — and is blended NRR lying?

AGE PROFILE · K3 · TENURE CROSS-SECTION

m60 ARR by age · 12-month display buckets of existing ages · logos ≠ ARR

8,959

0-11

8,869

12-23

8,780

24-35

8,693

36-47

27,625

48-60

48-60 includes the opening vintage. Tenure is flat — Scenario 6 / Tenure give age a rate.

RATES BY AGE · EDGE OF WINDOW

GRR solid · NRR dashed · gap = expansion (measured)

100%

90%

80%

70%

0-11

12-23

24-35

36-47

48-60

VINTAGE READING · WHICH VINTAGE CARRIES / LEAKS

m0

GRR 89.6% NRR 99.0%

leaks the book

expansion 9.4% · contraction —

m12

GRR 89.6% NRR 99.0%

leaks the book

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leaks the book

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THIS WORLD

Reading

Mode

Month 60 ARR · €000

R12M GRR

R12M NRR

Base

Age profile

62,926

89.6%

99.0%

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m0 share of book

m12 share of book

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30.2%

1.1%

1.2%

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Share of closing ARR at the selected month. Carry / leak is retained vs that vintage's own initial ARR.

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Scenario 6 · same KPIs, different history

Tenure laws on the rail write the same three-band array. No new coefficient.

Month 60 · year 5

SPEED

0.5x

1x

2x

4x

■ SaaS Physics an economic system, observed

base vs experiment

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Appendix

BASIS

MRR

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EXPERIMENT 0 assumptions changed · Experiment equals Base	Default world BaseExperimentCSV Same run as System. Money in €000. One world at a time.															APPENDIX
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	Month	▼ Revenue €000						▼ Retention		► Unit economics	▼ Capital €000				► Logos	
		ARR	MRR (NATIVE)	NEW ARR	EXPANSION ARR	LEAKAGE	NET NEW ARR	R12M GRR	R12M NRR		CASH	BURN	FCF	EBITA		
	0	20,000	1,667	—	—	—	—	—	—		10,000	—	—	—		
	1	20,733	1,728	750	158	175	733	—	—		9,408	592	-592	-592		
	2	21,466	1,789	750	164	181	733	—	—		8,864	543	-543	-543		
	3	22,198	1,850	750	170	188	732	—	—		8,370	495	-495	-495		
	4	22,929	1,911	750	175	194	731	—	—		7,924	446	-446	-446		
	5	23,660	1,972	750	181	200	731	—	—		7,527	397	-397	-397		
	6	24,390	2,033	750	187	207	730	—	—		7,179	348	-348	-348		
	7	25,120	2,093	750	193	213	730	—	—		6,879	300	-300	-300		
	8	25,849	2,154	750	199	220	729	—	—		6,628	251	-251	-251		
	9	26,577	2,215	750	204	226	728	—	—		6,426	202	-202	-202		
	10	27,305	2,275	750	210	232	728	—	—		6,272	154	-154	-154		
	11	28,032	2,336	750	216	239	727	—	—		6,166	105	-105	-105		
	12	28,759	2,397	750	222	245	727	89.56%	99.00%		6,109	57	-57	-57		
	13	29,485	2,457	750	227	251	726	89.56%	99.00%		6,101	9	-9	-9		
	14	30,210	2,517	750	233	258	725	89.56%	99.00%		6,141	0	40	40		
	15	30,935	2,578	750	239	264	725	89.56%	99.00%		6,229	0	88	88		
	16	31,659	2,638	750	245	270	724	89.56%	99.00%		6,365	0	136	136		
	17	32,382	2,699	750	250	277	723	89.56%	99.00%		6,550	0	185	185		
	18	33,105	2,759	750	256	283	723	89.56%	99.00%		6,783	0	233	233		
	19	33,827	2,819	750	262	289	722	89.56%	99.00%		7,064	0	281	281		
	20	34,549	2,879	750	267	296	722	89.56%	99.00%		7,393	0	329	329		
	21	35,270	2,939	750	273	302	721	89.56%	99.00%		7,770	0	377	377		
	22	35,991	2,999	750	279	308	720	89.56%	99.00%		8,196	0	425	425		
	23	36,710	3,059	750	284	315	720	89.56%	99.00%		8,669	0	473	473		
	24	37,430	3,119	750	290	321	719	89.56%	99.00%		9,190	0	521	521		
	25	38,148	3,179	750	296	327	719	89.56%	99.00%		9,760	0	569	569		
	26	38,866	3,239	750	302	333	718	89.56%	99.00%		10,377	0	617	617		
	27	39,584	3,299	750	307	340	717	89.56%	99.00%		11,042	0	665	665		
	28	40,301	3,358	750	313	346	717	89.56%	99.00%		11,755	0	713	713		
	29	41,017	3,418	750	319	352	716	89.56%	99.00%		12,515	0	761	761		
	30	41,733	3,478	750	324	359	716	89.56%	99.00%		13,324	0	808	808		
	31	42,448	3,537	750	330	365	715	89.56%	99.00%		14,180	0	856	856		
	32	43,162	3,597	750	336	371	714	89.56%	99.00%		15,083	0	904	904		
	33	43,876	3,656	750	341	377	714	89.56%	99.00%		16,035	0	951	951		